CO1 AT2 ANALYTICAL SOLVING

SIMATS Engineering • CSE • CSA0804 — Python Programming • 24-07-2026 • Category: Analytical • Student Submissions Report

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Q1. Harshad number

ANSWER

START
READ N
CALCULATE digit sum
IF N % digit_sum == 0
DISPLAY "Harshad number"
ELSE
DISPLAY "Not a Harshad Number"
PRINT all Harshad numbers from 1 to 500
STOP

Q2. GCD

CODE

```
a = int(input("Enter first: "))
b = int(input("Enter second: "))
while b != 0:
    temp = a
    a = b
    b = temp%b
    print("GCD is",a)
```

INPUT

```
24
36
```

OUTPUT

```
Enter first: Enter second: GCD is 36
GCD is 24
GCD is 12
```

Q3. sum to a target value

ANSWER

START
READ list and target
CREATE seen and pairs
CHECK each number
IF complement exists
STORE pair
ADD number to seen
PRINT pairs
STOP

Q4. Bitwise Operators

ANSWER

START
ASSIGN values to a,b, and c
CALCULATE result = (a AND b) OR c
DISPLAY a,b,c and result
DISPLAY result in binary, octal, and hexadecimal
DISPLAY result shifted right by 2, shifted left by 1, and bitwise complement
STOP

Q5. binary search

CODE

```
def rotated_binary_search(arr,target):
    low = 0
    high = len(arr) - 1
    while low ≤ high:
        mid = (low + high) // 2
        if arr[mid] == target:
            return mid
        if arr[low] ≤ arr[mid]:
            if arr[low] ≤ target < arr[mid]:
                high = mid - 1
            else:
                low = mid + 1
         else:
            if arr[low] ≤ arr[mid]:
                low = mid + 1
                else:
                    high = mid - 1
     return - 1
data = [4,5,6,7,0,1,2]
t = int(input())
result = rotated_binary_search(data,t)
print(result)
```

OUTPUT

```
  File "C:\Windows\TEMP\sandbox_192521375RD_6a624fc464c344.36404938\main.py", line 13
    else:
         ^
IndentationError: unindent does not match any outer indentation level
```